READTRACK: PERSONAL READING PROGRESS TRACKER

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY
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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "ReadTrack: Personal Reading Progress Tracker" is the bonafide work of **Pavithra Narayanan (241401064)**, who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year [2026–2027], Semester [V], under my supervision.

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DECLARATION

I, **Pavithra Narayanan (241401064)**, hereby declare that the project report titled "ReadTrack: Personal Reading Progress Tracker", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report, including its data model, forms, reports, pages, Deluge scripts, workflows and configuration, was designed, built and tested by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

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PAVITHRA NARAYANAN

ABSTRACT

Readers who keep several books going at once usually track them in notebooks, notes apps or spreadsheets. Book details, progress notes and reviews end up in different places, progress has to be counted by hand, dates and page numbers are not checked, nothing confirms what has been logged, and there is no control over who can see or approve what. Existing reading platforms offer shelves and reviews but fixed fields, no personal automation and little control over access.

ReadTrack is a personal reading progress tracker built on Zoho Creator, a low-code platform chosen for its fast development, easy maintenance by a non-specialist, built-in hosting and security, portal access and low cost. The application has three related forms (My Books, Reading Progress and Book Reviews) and a Reading Session subform inside My Books. Reports, a calendar report, a pivot table, a dashboard and a home page present the data, and a custom Goal Tracker widget built with HTML, CSS and JavaScript shows progress against a page goal. Deluge scripts validate the start date, pages read and update date, and fill the update date automatically. Email workflows confirm new books, reviews and progress entries, a review approval process lets the administrator approve or reject reviews, a blueprint controls how a book moves between reading statuses, and a batch workflow processes progress records. New book records are synchronised to a Zoho Sheet through a connection. Access is controlled through Admin and Reader permission sets and sharing rules that give the administrator read-only access to reader records.

The application was tested with sample data, the role-based access behaviour was confirmed for the Admin and Reader roles, and the application was deployed. The Goal Tracker widget works on its own inputs and is not yet connected to the stored progress records, which is noted as a limitation. The project shows how a small but complete application with automation, approval, integration and access control can be built on a low-code platform.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Zoho Sheet Integration; Role-Based Access Control

PROJECT AT A GLANCE

Attribute	Details
Project Title	ReadTrack: Personal Reading Progress Tracker
Application Domain	Education / Personal productivity (reading management)
Student Name & Register No.	Pavithra Narayanan – 241401064
Zoho Creator Edition	Trial
Application Name (Link Name)	ReadTrack
Published / Live URL	https://readtrack.zohocreatorportal.in
Demo Credentials (per role)	[Admin: Pavithra Narayanan | Reader: redjohn]
Number of Forms / Subforms	3 / 1 (Reading Session in My Books)
Number of Reports (by type)	List: 3 (All My Books, All Progress, All Book Reviews); Calendar: 1; Pivot: 1 (Zoho Analytics);
Number of Pages / Dashboards	1 dashboard (ReadTrack Dashboard), 1 home page (with Goal Tracker widget)
Number of Workflows	Form: 5 (4 validation / automatic-entry scripts, 1 Zoho Sheet sync); Email: 3; Approval: 1; Blueprint: 1; Batch: 1;
Deluge Functions / Custom Actions	0 standalone functions; 1 custom action on All My Books; 1 custom widget
Integrations	Zoho Sheet (connection, working); Zoho Analytics (pivot table); email notifications
Roles & Profiles Defined	Roles: Admin, Reader; permission sets: Admin Access, Reader Access; 3 sharing rules
Environments Used	Development; published as a live application
Date of Production Deployment	31 October 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CO	Course Outcome
CSS	Cascading Style Sheets
ER	Entity–Relationship
FR	Functional Requirement
HTML	HyperText Markup Language
IEEE	Institute of Electrical and Electronics Engineers
NFR	Non-Functional Requirement
OAuth	Open Authorization
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator
Zia	Zoho's AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Reading is a long-term habit for students and professionals, but keeping track of it is usually informal. A reader often has several books open at the same time, remembers roughly where each one stands, and rarely records an opinion once a book is finished. Popular services such as Goodreads [1] and The StoryGraph [2] offer shelves, progress updates and reviews. They are, however, general-purpose platforms: a reader cannot change the data fields, add personal automation, or decide exactly which records are visible to whom.

This project builds **ReadTrack: Personal Reading Progress Tracker** on Zoho Creator. The "organisation" is an individual reader or a small reading group. Readers add books, log reading sessions and progress, and write reviews; an administrator approves reviews and can see the records that readers add. Reports, a dashboard and a home page show progress, and new books are copied to a Zoho Sheet.

1.2 Problem Statement

Readers who track their books in notebooks, notes apps or spreadsheets have no single place that connects a book with its progress and its review, and nothing confirms what has been logged. The same difficulty appears when a class, a reading club or a study group follows a shared reading list: every member keeps separate notes, progress cannot be compared, reviews cannot be moderated, and there is no way to share selected records without giving access to everything.

Although the domain is small, the underlying problem is a typical business-data problem. Related records must be linked, inputs must be checked, notifications and approvals must happen at the right time, and access must be limited by role. These are exactly the problems that low-code business applications are built to solve, so ReadTrack uses a familiar everyday domain to show them end to end. The main pain points are:

- Book details, progress notes and reviews are stored in different places.
- Progress is not summarised automatically; the reader must count pages and statuses manually.

- Invalid data, such as a start date in the future or zero pages read, is not checked at the time of entry.
- There is no confirmation when a book, a progress entry or a review is added, and no approval step for reviews.
- There is no access control: everything is either private or shared completely.

1.3 Objectives

- To design a structured data model with three related forms and a subform for books, reading progress, reading sessions and book reviews.
- To automate data checks, automatic date entry and email notifications using Deluge scripting and workflows.
- To provide reports, a calendar report, a pivot table, a dashboard and a home page that show book records and reading statistics.
- To control the reading status of a book with a blueprint, and to approve book reviews with an approval process.
- To integrate the application with Zoho Sheet so that new book records are synchronised automatically.
- To control access through Admin and Reader permission sets, roles and sharing rules, and to extend the home page with a custom widget.
- To test, deploy and govern the application with security controls.

1.4 Scope of the Project

In scope: a web application on Zoho Creator with three forms (My Books, Reading Progress, Book Reviews) and a Reading Session subform; three list reports, a calendar report and a pivot table; a dashboard and a home page; four validation and automatic-entry scripts; three email workflows; an approval process; a blueprint; a batch workflow; a Zoho Sheet integration; a custom Goal Tracker widget; and Admin and Reader permission sets with sharing rules.

Out of scope: dedicated mobile layouts, payment features, automatic book lookup from external catalogue APIs, social features such as following other readers, and connecting the Goal Tracker widget to the stored progress records.

1.5 Justification for a Low-Code Approach

Low-code platforms let applications be built mainly through visual configuration with only a small amount of hand-written code [3], although practitioners report challenges, so suitability should be checked for each project [4]. Zoho Creator was chosen for ReadTrack for the following reasons.

- **Development speed:** forms, reports and workflows are configured visually, so a working application can be produced in much less time than a coded one [3].
- **Maintainability by a citizen developer:** the owner can add a field, change a report or adjust a rule in the builder or in a short Deluge script, without a development team.
- **Built-in hosting:** the platform hosts the application and stores its data, so no server, database or deployment pipeline has to be set up [5].
- **Mobile and portal access:** Zoho Creator applications can be opened in the Zoho Creator mobile app, and portal access lets readers sign in without a full user licence set-up [5].
- **Security:** sign-in, roles, permission sets and sharing rules are provided by the platform, so access control is configured rather than coded [5].
- **Cost:** there is no separate cost for servers or for a development team; the main cost is the platform subscription, which is small for a personal-scale application.

The limitations met in this project are discussed in Section 8.3.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, security and governance. Chapter 8 presents the results. Chapter 9 gives the conclusion and future enhancements. The appendices contain the feature checklist, course outcome mapping, Deluge source code and user manual.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

At present, a reader tracks books informally. The reader picks a book, notes the start date in a notebook, jots down pages read in a notes app after each session, and later writes a review somewhere else, if at all. To see overall progress, the reader has to search through these separate notes by hand. Figure 2.1 shows this flow.

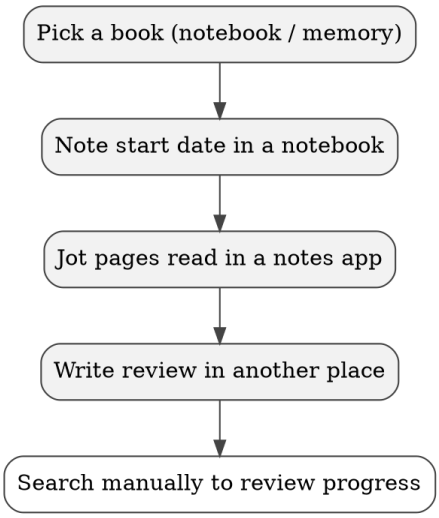


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Data is scattered across notebooks, notes and other tools, so nothing links a book to its progress and review.
- Progress and status counts are calculated manually and are often out of date.
- There is no validation of dates or page numbers at the time of entry.
- There is no confirmation or notification when something is logged, and reviews cannot be approved.
- Records cannot be shared selectively or in read-only mode.

2.3 Low-Code versus Traditional Development

Table 2.1 compares the two approaches for ReadTrack.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Estimated 4–6 weeks: database design, web screens, login, hosting, testing	Estimated 1–2 weeks: forms, reports and workflows configured visually
Skills required	Web framework, database, server administration, security	Basic data modelling and short Deluge scripts
Hosting & infrastructure	Must be arranged and maintained separately	Provided by the platform
Mobile / portal access	Separate mobile build or responsive redesign	Zoho Creator mobile app and portal access available
Security & access control	Login, roles and permissions coded by hand	Permission sets, roles and sharing rules are configured
Maintenance & change	Code change, rebuild and redeploy	Edit form or workflow and publish
Customisation limits	Almost unlimited	Limited to platform features, Deluge and widgets
Cost	Developer time and hosting cost	Platform subscription; low for personal scale

2.4 Literature Survey

Eight sources were reviewed: two existing reading-tracker products, four academic or scholarly works on low-code development, and the official Zoho documentation. Table 2.2 summarises them.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Goodreads [1]	Goodreads	Social platform where readers keep shelves, update progress and write reviews	Shows the common features of a reading tracker; fixed fields and no personal automation
2	The StoryGraph [2]	The StoryGraph	Reading tracker with progress logging and reading statistics	Confirms the demand for progress statistics; ReadTrack makes them configurable
3	A. C. Bock and U. Frank, 2021 [3]	Low-Code Platform	Describes what low-code platforms are and how they support application building	Basis for the low-code justification in Section 1.5

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
4	Y. Luo et al., 2021 [4]	Characteristics and Challenges of Low-Code Development: The Practitioners' Perspective	Studied Stack Overflow and Reddit discussions to identify characteristics and challenges of low-code development; advises checking whether it fits the project	Guides the limitations discussion in Section 8.3
5	Zoho Corporation [5]	Zoho Creator Help Documentation	Explains forms, reports, pages, workflows, connections, widgets and permissions	Primary reference for the implementation in Chapter 5
6	A. Sahay et al., 2020 [6]	Supporting the understanding and comparison of low-code development platforms	Supports understanding and comparison of low-code platforms	Supports the choice of a suitable platform for the application
7	R. Waszkowski, 2019 [7]	Low-code platform for automating business processes in manufacturing	Describes the use of a low-code platform to automate business processes	Shows that process automation is a core low-code use; applied here to reading workflows
8	Zoho Corporation [8]	Deluge Script Reference Guide	Describes Deluge syntax, tasks and functions	Reference for all scripts in Appendix C

2.5 Emerging Trends in Low-Code Development

Four trends influenced the design. **AI-assisted building:** platforms such as Zoho Creator offer Zia, an AI assistant that can help start an application from a plain-language description; in this project it was used only for a few suggestions and for help with some Deluge scripts. **Citizen development:** the application is small enough to be built and maintained by a non-specialist. **Hyper-automation:** email notifications, an approval process, a blueprint and a Zoho Sheet synchronisation remove manual follow-up. **Composable applications:** Zoho Sheet and Zoho Analytics are reused instead of building a spreadsheet export or a pivot engine from scratch.

2.6 Proposed System

ReadTrack keeps books, progress and reviews in three related forms, so each progress update or review is linked to its book through a lookup. Validation scripts and automatic date entry keep the data clean (Limitation 3). Reports, a pivot table and a dashboard summarise progress without manual counting (Limitation 2). Email workflows, an approval process and a blueprint give confirmation and control (Limitation 4).

Permission sets and sharing rules provide controlled, read-only sharing (Limitation 5). All data is held in one application instead of scattered notes (Limitation 1).

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator (Admin Access)	Owner of the application	Manage permissions and sharing, approve or reject reviews, view the records that readers add (read-only), use the dashboard and home page
Reader (Reader Access)	Person who reads and records books	Add books, log reading sessions and progress, write reviews; works only with own records

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow a user to add a book with title, author, genre, start date and status (mandatory), and optional ISBN, personal notes, favourite flag and reference link, and to log reading sessions in a subform.	Admin, Reader	High
FR-02	The system shall record reading progress (pages read, date updated, chapter read, notes) for a book selected through a lookup.	Admin, Reader	High
FR-03	The system shall record a rating (1–5), review and review date for a book selected through a lookup.	Admin, Reader	Medium
FR-04	The system shall reject a book start date that is in the future and cancel the submission.	Admin, Reader	High
FR-05	The system shall reject a progress record whose pages read is zero or negative, or whose update date is in the future.	Admin, Reader	High
FR-06	The system shall set the update date to today's date when it is left empty.	Admin, Reader	Medium
FR-07	The system shall send an email when a new book, a progress record or a review is added.	Admin, Reader	Medium
FR-08	The system shall send each submitted review to the administrator, who can approve or reject it.	Admin, Reader	High
FR-09	The system shall control book status changes through a blueprint (To Read, Currently Reading, Completed, On Hold).	Admin, Reader	Medium
FR-10	The system shall process Reading Progress records through a batch workflow.	Admin	Low

FR ID	Requirement	Role(s)	Priority
FR-11	The system shall copy each new book record to a Zoho Sheet.	Admin, Reader	Medium
FR-12	The system shall provide the reports All My Books, All Progress and All Book Reviews, a calendar report and a pivot table.	Admin	High
FR-13	The system shall provide a dashboard with book counts and charts, and a home page with a goal tracker widget.	Admin	Medium
FR-14	The system shall restrict access by role: Admin and Reader permission sets, with sharing rules that give the Admin read-only access to reader records.	All	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports for a personal library (up to a few hundred records) shall open within a few seconds.
NFR-02	Security	Users shall access only the data permitted by their permission set and sharing rules.
NFR-03	Usability	Forms shall be simple, with clear labels and validation messages; the application shall work in a standard web browser or through the portal.
NFR-04	Availability	The application shall be available whenever the Zoho Creator service is available.
NFR-05	Maintainability	Scripts shall be short, commented and named after the workflow they belong to.
NFR-06	Auditability	Record changes shall be traceable through the platform's audit trail.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Widget technologies	HTML, CSS and JavaScript (ReadTrack Goal Tracker widget)
Browser	Google Chrome
Other Zoho services	Zoho Sheet, Zoho Analytics

Category	Specification
Access	Web browser and portal access; Zoho Creator mobile app not used in this project
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

Figure 3.1 shows the use cases for the two roles. Tables 3.5 and 3.6 describe two key use cases.

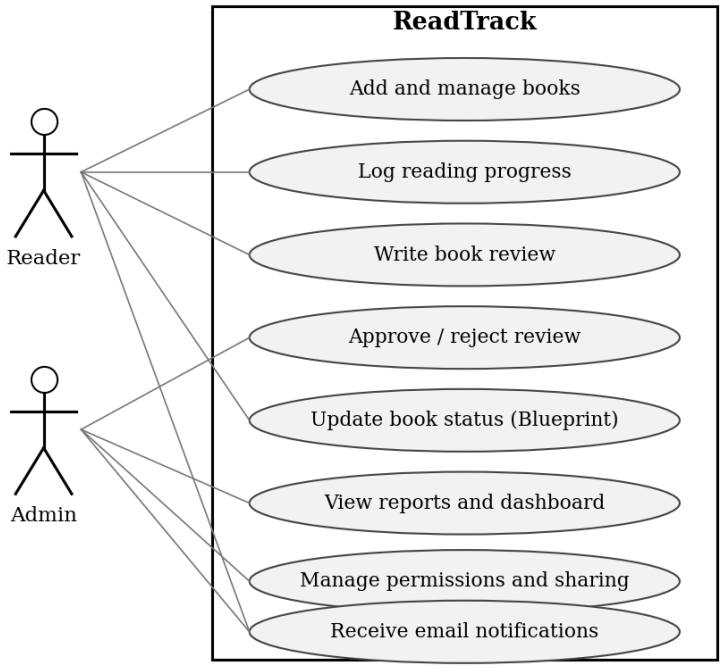


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Add a Book

Item	Description
Use Case ID	UC-01
Use Case Name	Add a book
Actor(s)	Reader, Admin
Pre-condition	User has a Zoho account and Reader or Admin access
Main Flow	1. User opens the My Books form. 2. User enters title, author, genre, start date and status, and optional details. 3. User may add reading sessions in the subform. 4. User submits the form. 5. A new-book email is sent and the book is copied to the Zoho Sheet.

Item	Description
Alternate Flow	If the start date is in the future, an alert is shown and the submission is cancelled. If a mandatory field is empty, the form is not saved.
Post-condition	The book appears in the All My Books report and in the Zoho Sheet
Related FR	FR-01, FR-04, FR-07, FR-11

Table 3.6: Use Case Description – Submit a Book Review

Item	Description
Use Case ID	UC-02
Use Case Name	Submit a book review
Actor(s)	Reader (submits), Admin (approves)
Pre-condition	At least one book exists in My Books
Main Flow	1. Reader opens the Book Reviews form. 2. Reader selects a book and enters rating, review and review date. 3. Reader submits the form. 4. A review email is sent and the Admin receives an approval notification. 5. Admin approves or rejects the review.
Alternate Flow	If the Admin rejects the review, it is not approved and the Reader may submit a corrected review.
Post-condition	Approved review appears in the All Book Reviews report
Related FR	FR-03, FR-07, FR-08, FR-12

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

Figure 4.1 shows the layers of ReadTrack. Users sign in through a web browser or the portal and pass through role-based access. The Zoho Creator application contains the forms, the reports, the dashboard and the home page with its widget, and the Deluge scripts and workflows. Form records are stored in the platform's data layer. New book records are sent to Zoho Sheet through an authorised connection, and the pivot table is built with Zoho Analytics.

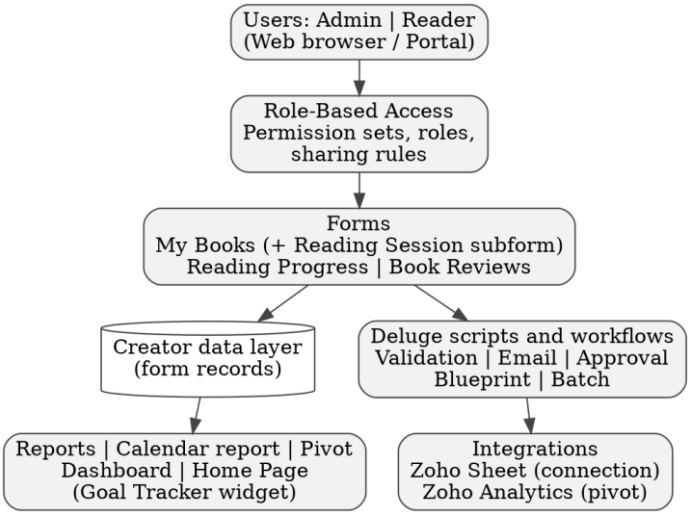


Figure 4.1: System Architecture of the Application

4.2 Data Model

The data model has three forms and one subform (Figure 4.2). **My Books** is the master form and contains the **Reading Session** subform for individual reading sessions. **Reading Progress** and **Book Reviews** each refer to one book through a Lookup field, so one book can have many progress updates and many reviews (one-to-many).

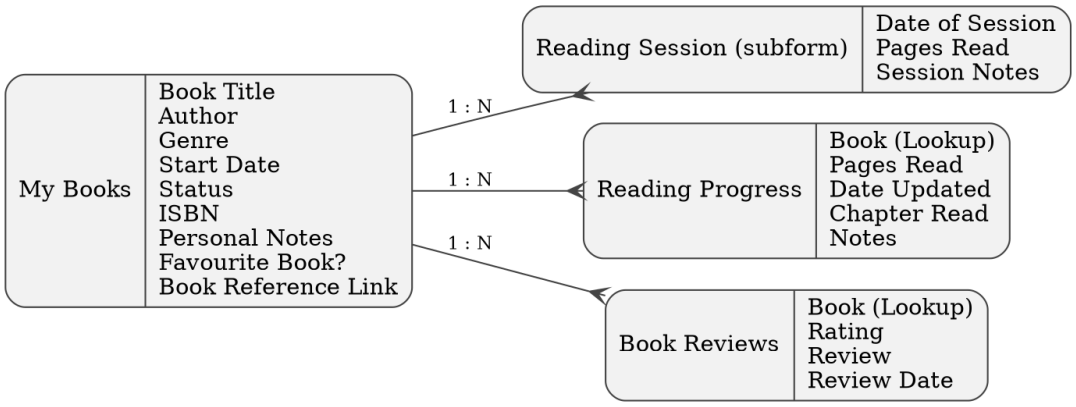


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
My Books	Book Title, Author, Genre, Start Date, Status	Reading Session	One-to-Many	Subform inside My Books
My Books	Book Title, Author, Genre, Start Date, Status	Reading Progress	One-to-Many	Lookup (Book) in Reading Progress
My Books	Book Title, Author, Genre, Start Date, Status	Book Reviews	One-to-Many	Lookup (Book) in Book Reviews
Reading Progress	Book, Pages Read, Date Updated	My Books	Many-to-One	Lookup field
Book Reviews	Book, Rating, Review Date	My Books	Many-to-One	Lookup field

4.3 Form Design

Table 4.2: Field Specification – My Books Form

Field Label	Field Type	Mandatory	Validation / Default	Options	Remarks
Book Title	Single Line	Yes	—	—	—
Author	Single Line	Yes	—	—	—
Genre	Dropdown	Yes	—	Fantasy, Mystery, Romance, Science Fiction, Non-fiction, Thriller, Historical Fiction, Self-Help, Literary Fiction	—
Start Date	Date	Yes	Not in the future	—	Checked by script DS-01
Status	Dropdown	Yes	—	Want to Read, Reading, Completed, On Hold	Used by dashboard and blueprint
ISBN	Single Line	No	—	—	—

Field Label	Field Type	Mandatory	Validation / Default	Options	Remarks
Personal Notes	Multi Line	No	—	—	—
Favourite Book?	Checkbox	No	—	—	Shown in All My Books report
Book Reference Link	URL	No	—	—	—

Table 4.3: Field Specification – Reading Session Subform (inside My Books)

Field Label	Field Type	Mandatory	Remarks
Date of Session	Date	No	Date of the reading session
Pages Read	Number	No	Pages read in that session
Session Notes	Multi Line	No	Notes about the session

Table 4.4: Field Specification – Reading Progress Form

Field Label	Field Type	Mandatory	Validation / Default	Remarks
Book	Lookup (My Books)	Yes	—	Selects the book
Pages Read	Number	Yes	Greater than 0	Checked by script DS-02
Date Updated	Date	No	Not in the future; today if empty	Scripts DS-03 and DS-04
Chapter Read	Single Line	No	—	—
Notes	Multi Line	No	—	—

Table 4.5: Field Specification – Book Reviews Form

Field Label	Field Type	Mandatory	Validation / Default	Remarks
Book	Lookup (My Books)	Yes	—	Selects the book
Rating	Dropdown (1–5)	Yes	—	Star rating
Review	Multi Line	No	—	Written review
Review Date	Date	Yes	—	—

4.4 Process and Workflow Design

Figure 4.3 shows how a record moves through the automated process. When a form is submitted, the validation scripts run; invalid data stops the submission with an alert. If the update date of a progress record is empty, it is set to today's date. After the record is saved, the matching email workflow runs. A new book is copied to the Zoho Sheet,

a review is sent to the Admin for approval, and the status of a book moves only through the blueprint transitions.

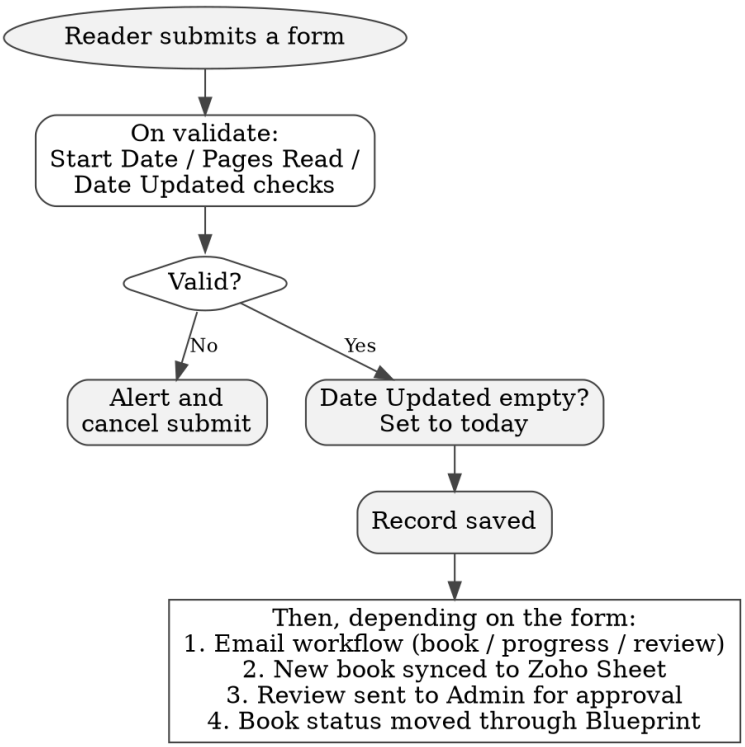


Figure 4.3: Workflow / Process Flow for Record Submission

4.5 User Interface and Navigation Design

The application menu has four groups: the home page, the dashboard, the reports and the forms (Figure 4.4). The home page holds a welcome and navigation section, the All Progress report and the Goal Tracker widget. The dashboard shows book counts and charts. The home page and dashboard are hidden from the Reader role because the shared statistics are not suitable for readers.

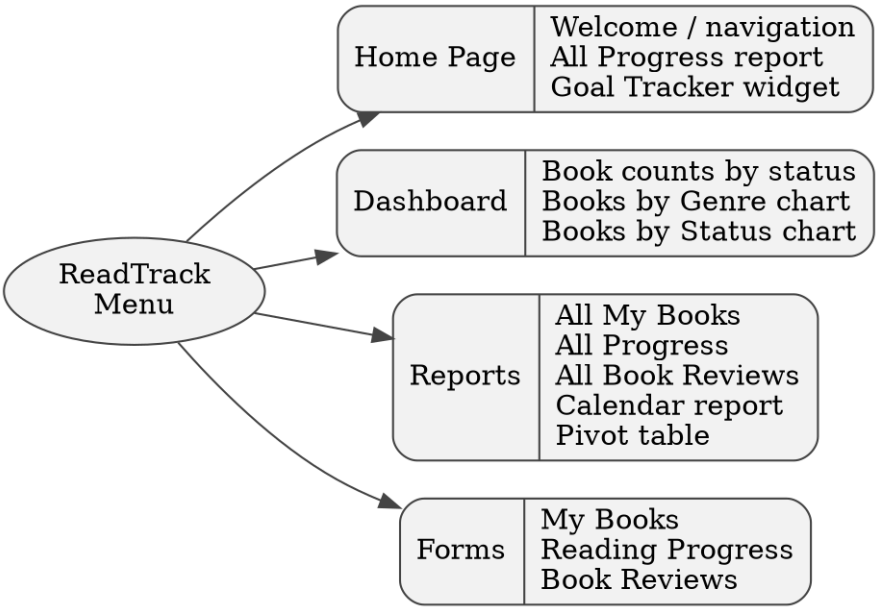


Figure 4.4: Navigation Structure and Wireframe of the Application

4.6 Access Control Design

Table 4.6 is the permission design implemented in Section 5.8. A = Add, E = Edit, V = View, D = Delete, — = No access. Reader Access has Access, View and Add permissions only, without View All or Modify All. Sharing rules give the Admin read-only access to the records that Readers add.

Table 4.6: Role–Permission Matrix

Component	Admin Access	Reader Access
Form: My Books	A E V D; reader records read-only	A V (own records)
Form: Reading Progress	A E V D; reader records read-only	A V (own records)
Form: Book Reviews	A E V D; approves / rejects	A V (own records); submits for approval
Reports	V	V (own records)
Dashboard / Home Page	V	Hidden
Settings and sharing	Full	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

A new application named **ReadTrack** was created in Zoho Creator. Zia, Zoho's AI assistant, was used only in a small way, to ask for suggestions and for help with a few Deluge scripts. The application, including its forms, reports, pages, workflows, integrations and permissions, was built and configured in Zoho Creator as described in this chapter. The menu groups the home page, the dashboard, the forms and the reports (Figure 5.1). Demo data of about 20 books, 10 progress entries and 6 reviews was added for testing.

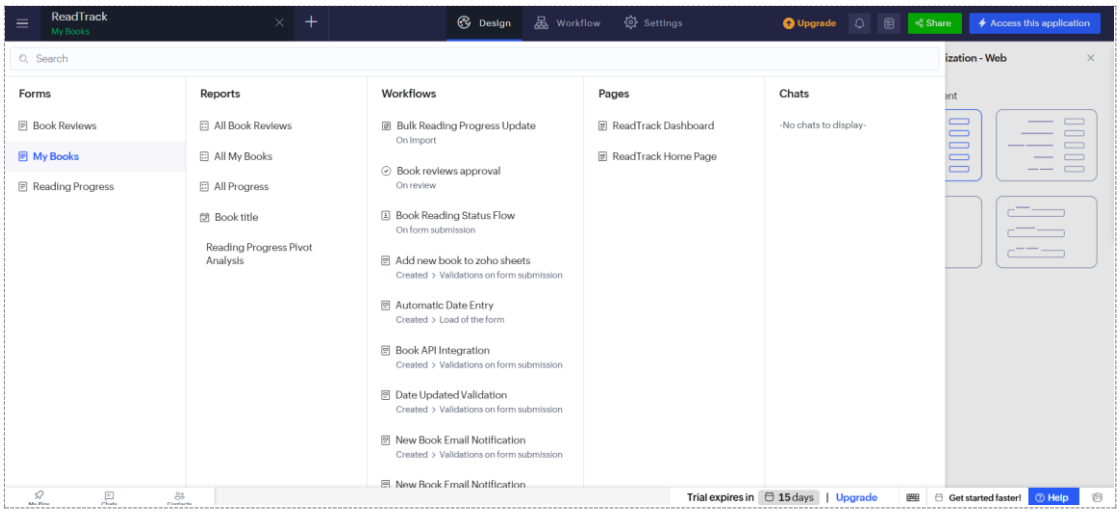


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

The application has three forms and one subform. **My Books** stores Book Title, Author, Genre, Start Date, Status, ISBN, Personal Notes, Favourite Book? and Book Reference Link. Book Title, Author, Genre, Start Date and Status are mandatory. Genre is a dropdown (Fiction, Fantasy, Mystery, Romance, Science Fiction, Non-fiction, Literary Fiction, Thriller, Historical Fiction, and Self-help) and Status is a dropdown (Want to Read, Reading, Completed, On Hold). The **Reading Session** subform inside My Books stores the Date of Session, Pages Read and Session Notes for each session. **Reading Progress** stores the selected book, pages read, date updated, chapter read and notes. **Book Reviews** stores the selected book, rating (1–5), review and review date. In both Reading Progress and Book Reviews the Book field is a lookup to My Books. Field details are in Tables 4.2 to 4.5.

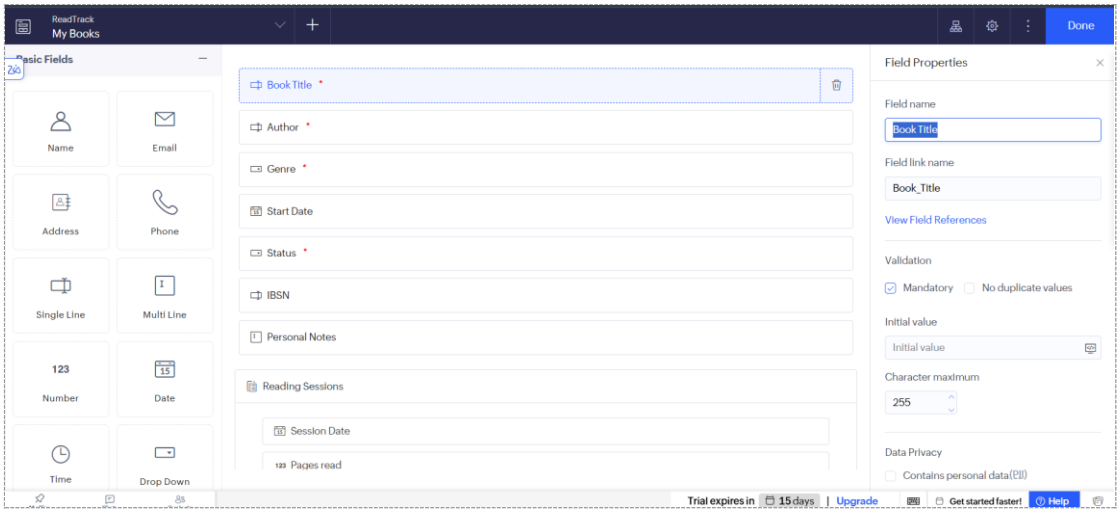


Figure 5.2: My Books Form with the Reading Session Subform

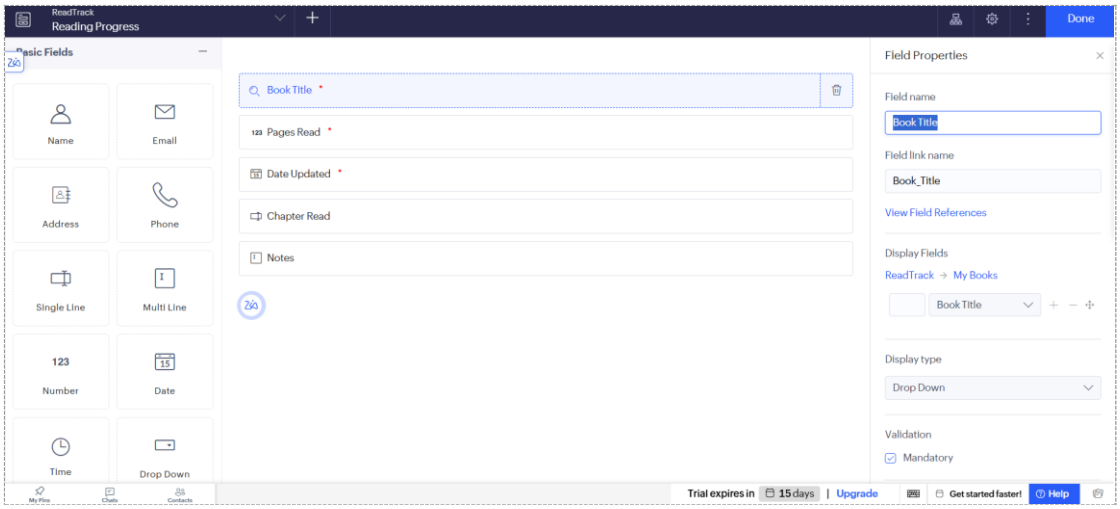


Figure 5.3: Reading Progress Form with the Book Lookup

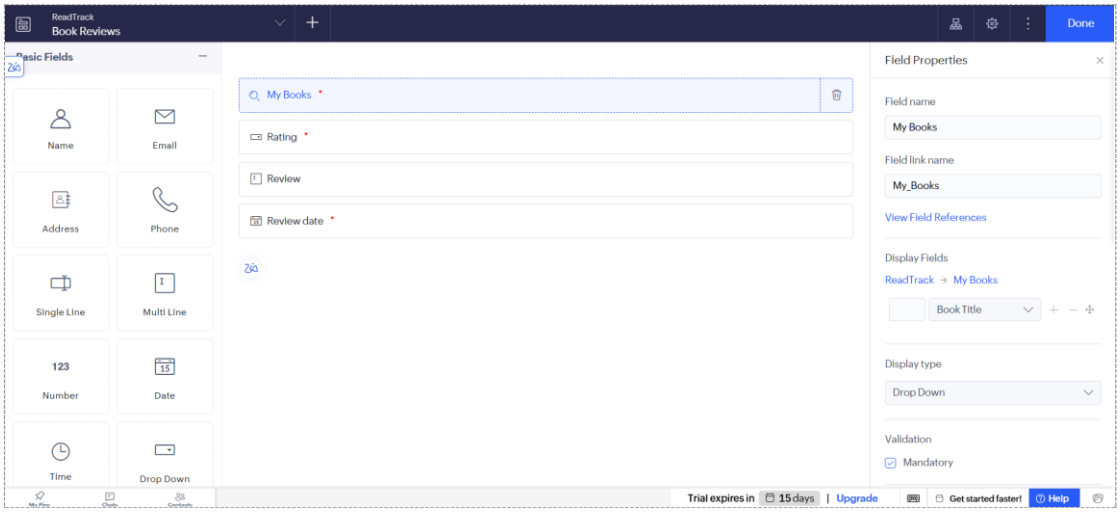


Figure 5.4: Book Reviews Form

5.3 Reports

Three list reports are available: All My Books, All Progress and All Book Reviews (Table 5.1). The All My Books report was customised with a Genre filter set to Fantasy, grouping by Author, the Favourite Book? checkbox field and a custom action. A calendar report was created from Reading Progress, with the book as the event name and Date Updated as both the start and end date. A pivot table was created with Zoho Analytics, with Book Title as the column and Pages Read as the row.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Display	Custom Actions	Roles
All My Books	List	My Books	Genre = Fantasy; grouped by Author; shows Favourite Book?	1 custom action	Admin; Reader (own records)
All Progress	List	Reading Progress	All records	None	Admin; Reader (own records)
All Book Reviews	List	Book Reviews	All records	None	Admin; Reader (own records)
Reading Progress Calendar	Calendar	Reading Progress	Event name: Book; start and end: Date Updated	None	Admin
Pivot Table (Zoho Analytics)	Pivot	Reading Progress	Column: Book Title; row: Pages Read	None	Admin

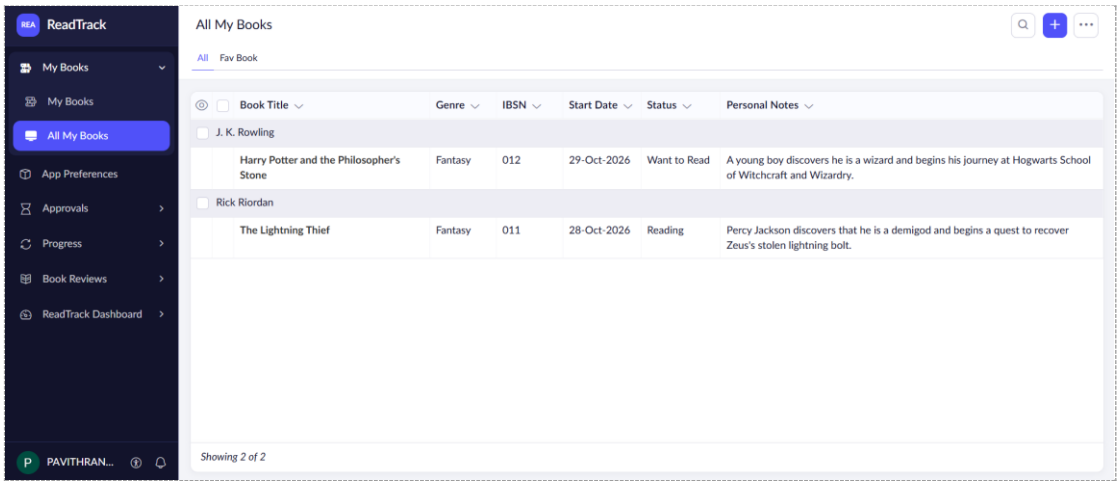


Figure 5.5: All My Books Report (Customised)

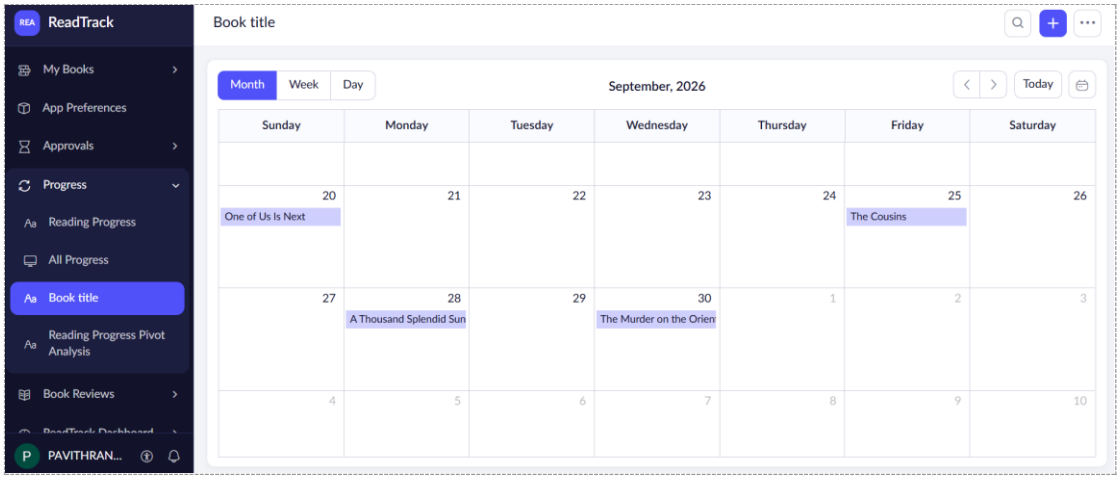


Figure 5.6: Reading Progress Calendar Report

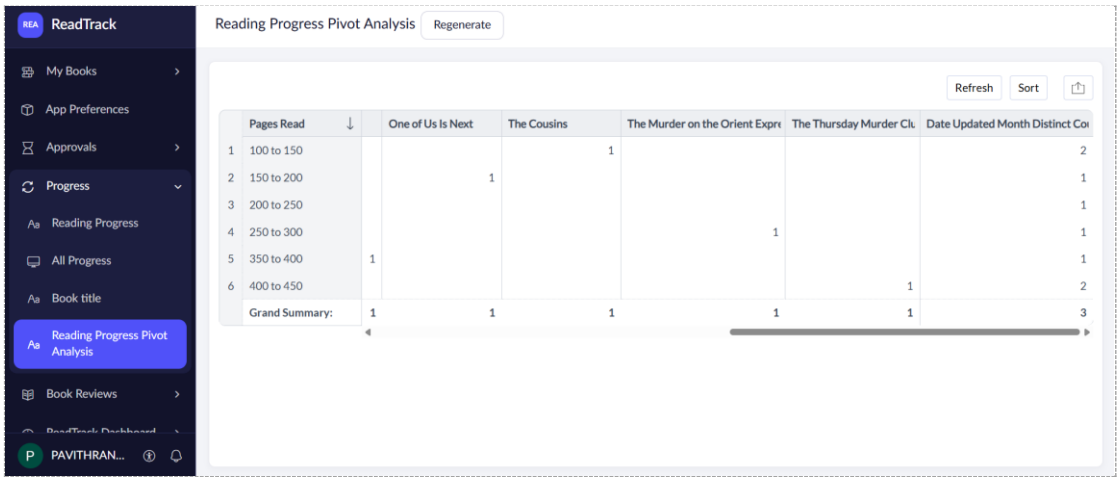


Figure 5.7: Pivot Table Created with Zoho Analytics

5.4 Pages and Dashboards

The **ReadTrack Dashboard** shows Total Books, Currently Reading, Completed Books and Books on Hold, together with a Books by Genre chart and a Books by Status chart. The **ReadTrack Home Page** contains a welcome and navigation section, the All Progress report and the ReadTrack Goal Tracker widget (Section 5.10). Both were previewed and compared with the stored data to confirm that the figures were correct.

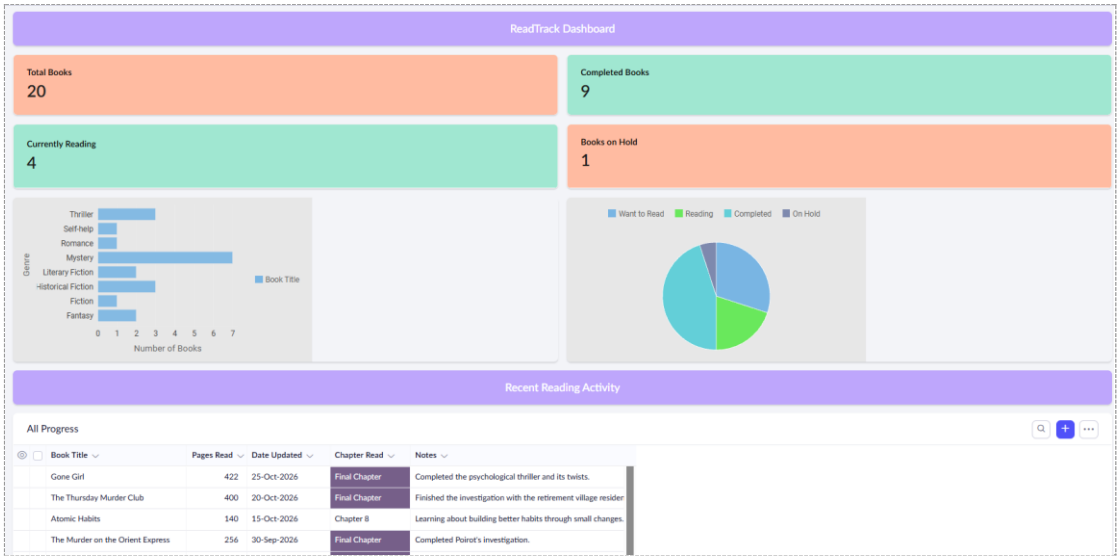


Figure 5.8: ReadTrack Dashboard

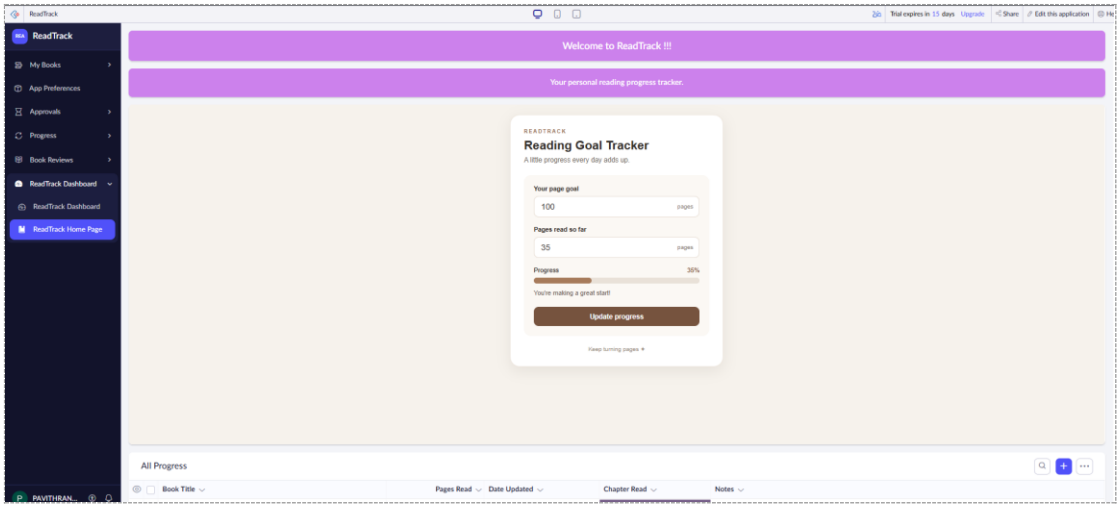


Figure 5.9: ReadTrack Home Page with the Goal Tracker Widget

5.5 Deluge Scripting

Five Deluge scripts support the form workflows (Table 5.2): three validations, one automatic date entry and one Zoho Sheet synchronisation. Codes 5.1 to 5.4 show the first four; the Zoho Sheet script is in Section 5.7 and the full set is in Appendix C. The scripts use if conditions, null checks, date comparison against the current date, alerts, cancelling a submission and the Zoho Sheet task with a connection. Zia was consulted only for suggestions and help with a few of these scripts; all scripts were checked and tested with sample records.

Table 5.2: Deluge Scripts Implemented

Script ID	Form / Event	Purpose	Deluge Concepts Used
DS-01	My Books → On validate	Reject a start date in the future	date comparison, alert, cancel submit
DS-02	Reading Progress → On validate	Reject Pages Read of zero or less	comparison, alert, cancel submit
DS-03	Reading Progress → On validate	Reject a Date Updated in the future	date comparison, alert, cancel submit
DS-04	Reading Progress → before submission	Set Date Updated to today when empty	isNull(), zoho.currentdate
DS-05	My Books → On success	Copy a new book to the Zoho Sheet	Map, zoho.sheet.createRecords, connection

Code 5.1: Start date validation (DS-01)

```
if(input.Start_Date > zoho.currentdate)
{
    alert "Start Date cannot be in the future.";
    cancel submit;
}
```

Code 5.2: Pages Read validation (DS-02)

```
if(input.Pages_Read <= 0)
{
    alert "Pages Read must be greater than zero. Please enter a
valid positive number.";
    cancel submit;
}
```

Code 5.3: Date Updated validation (DS-03)

```
if(input.Date_Updated > zoho.currentdate)
{
    alert "Date Updated cannot be in the future. Please select
today or an earlier date.";
    cancel submit;
}
```

Code 5.4: Automatic date entry (DS-04)

```
if(input.Date_Updated.isNull())
{
    input.Date_Updated = zoho.currentdate;
}
```

Result observed:

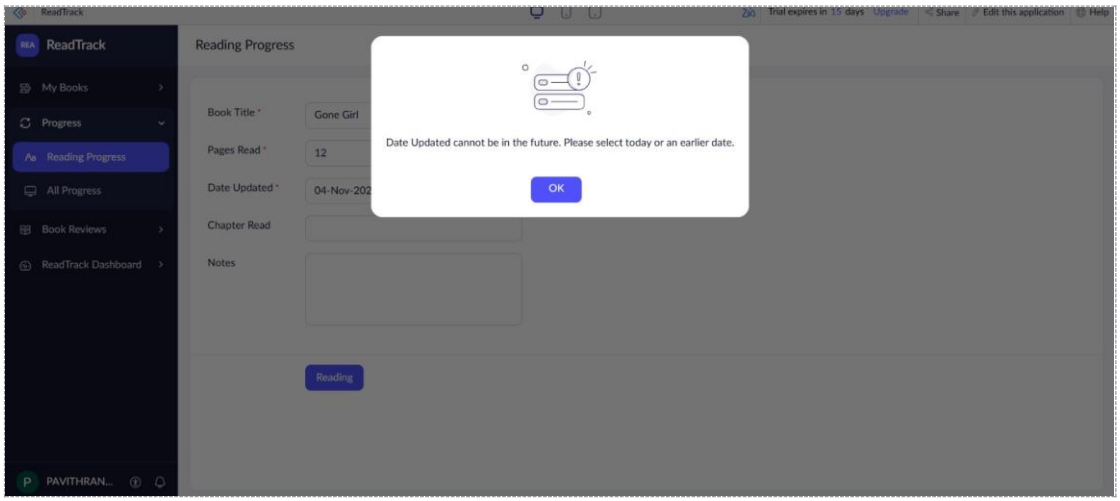


Figure 5.10: Deluge Code of the Validation Scripts

5.6 Workflows and Automation

Table 5.3 lists the workflows. Besides the form workflows that hold the scripts above, three email workflows, an approval process, a blueprint and a batch workflow were configured. There is no scheduled workflow because the current design does not need time-based actions.

Table 5.3: Workflows Implemented

Workflow Name	Type	Form	Trigger / Condition	Action(s)	Script Ref.
Start Date Validation	Form workflow	My Books	On validate; start date in the future	Alert and cancel submit	DS-01
Pages Read Validation	Form workflow	Reading Progress	On validate; pages read ≤ 0	Alert and cancel submit	DS-02
Date Updated Validation	Form workflow	Reading Progress	On validate; date in the future	Alert and cancel submit	DS-03
Automatic Date Entry	Form workflow	Reading Progress	Date Updated is empty	Set to today's date	DS-04
New Book Email	Email workflow	My Books	New book added	Email to the logged-in user	—
Reading Progress Email	Email workflow	Reading Progress	New progress added	Email to the logged-in user	—
Book Review Email	Email workflow	Book Reviews	New review added	Email to the logged-in user	—

Workflow Name	Type	Form	Trigger / Condition	Action(s)	Script Ref.
Zoho Sheet Sync	Form workflow	My Books	New book added	Create a record in the Zoho Sheet	DS-05
Review Approval	Approval process	Book Reviews	Reader submits a review	Admin approves or rejects	—
Book Reading Status Flow	Blueprint	My Books	Status changes	Allowed transitions only	—
Reading Progress Batch	Batch workflow	Reading Progress	Batch of up to 1000 records	Log "record processed"	—

Email workflows. New Book Email, Book Review Email and Reading Progress Email send a notification when a record is added to the matching form.

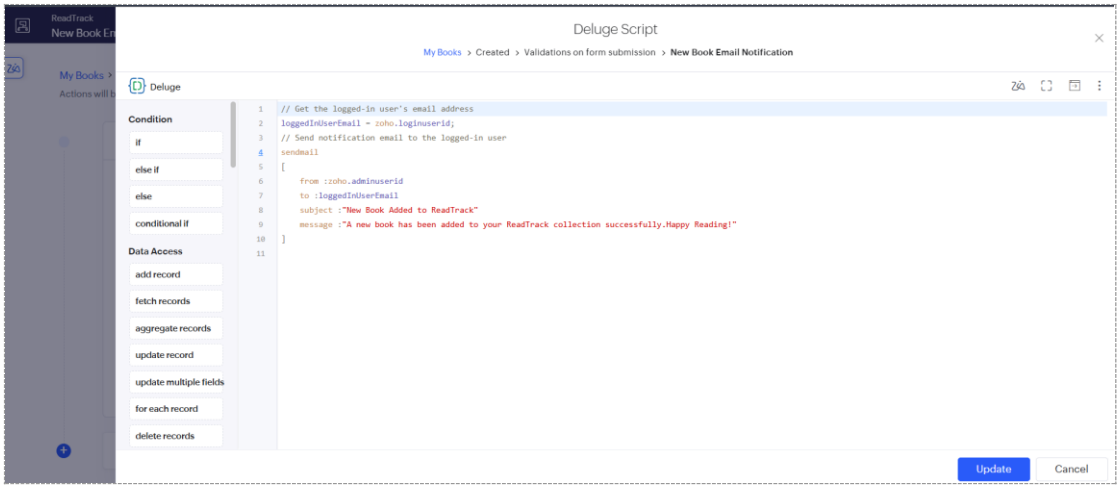


Figure 5.11: Email Workflows

Approval process. Book Reviews has an approval process. When a Reader submits a review, the Admin receives an approval notification and can approve or reject it.

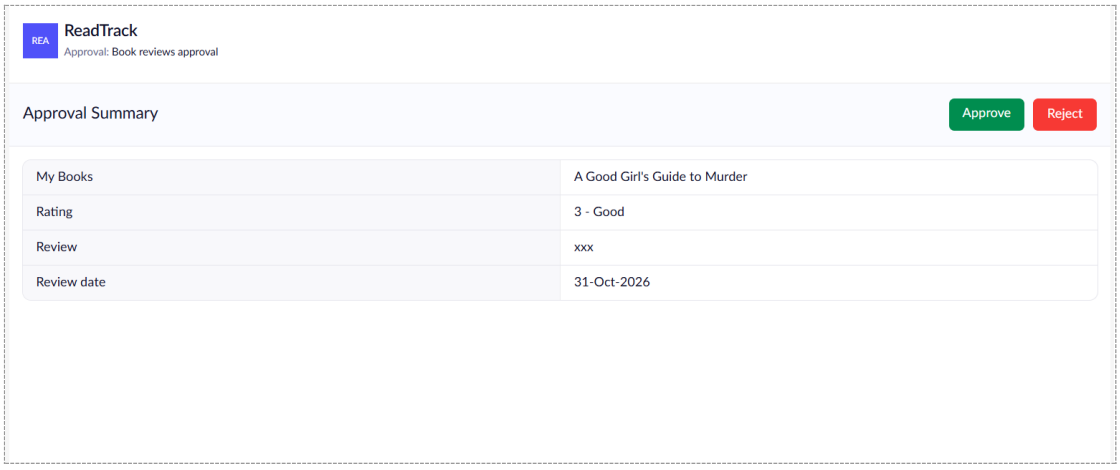


Figure 5.12: Approval Process for Book Reviews

Blueprint. The blueprint **Book Reading Status Flow** is attached to My Books and controls how a book moves between stages (Table 5.4). The blueprint defines only the allowed transitions between these statuses.

Table 5.4: Blueprint Stages and Transitions

From Stage	To Stage	Transition Name
To Read	Currently Reading	Start Reading
Currently Reading	Completed	Mark as Completed
Currently Reading	On Hold	Put on Hold
On Hold	Currently Reading	Resume Reading

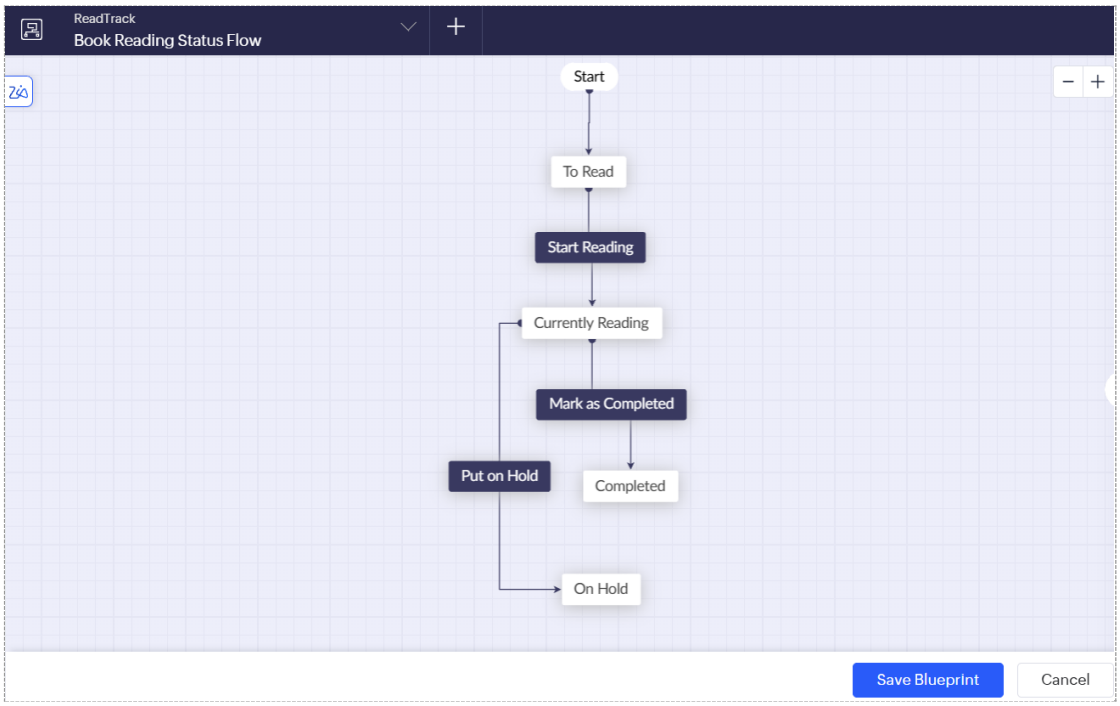


Figure 5.13: Book Reading Status Flow Blueprint

Batch workflow. A batch workflow is configured for Reading Progress with a batch size of 1000. For each batch it runs a script that prints the message "Reading Progress record processed successfully".

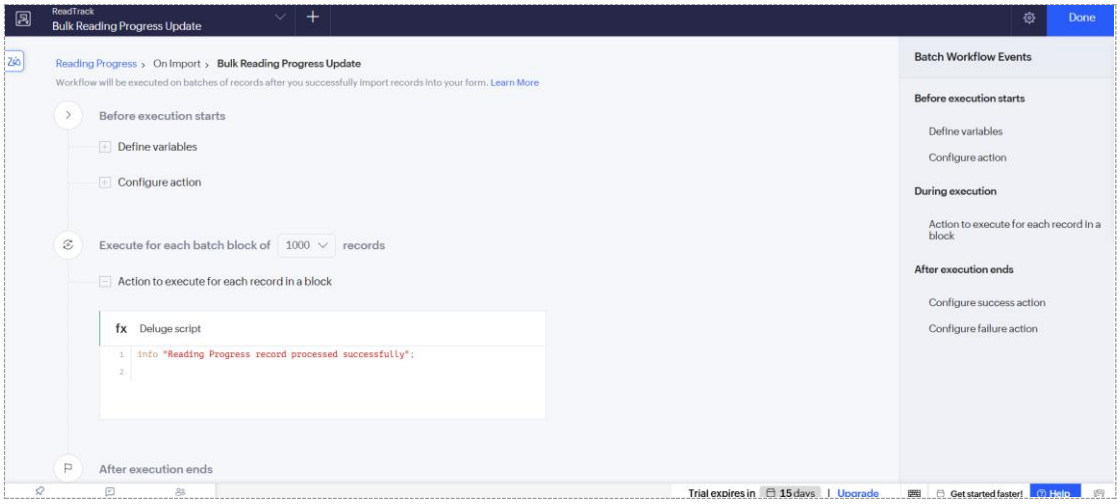


Figure 5.14: Batch Workflow for Reading Progress

5.7 Integrations

ReadTrack is integrated with **Zoho Sheet**. A connection named zoho_sheet_connection was created and authorised, and a form workflow uses it to copy each new book to the worksheet Sheet1 of the Zoho Sheet **ReadTrack Book Records**. Book Title, Author, Genre and Status are sent. The integration is working, and new book records successfully appear in the sheet. The connection uses OAuth, so no password or key is written in the script. The pivot table in Section 5.3 uses the Zoho Analytics integration.

Table 5.5: Integrations Implemented

Integration	Service / Resource	Method	Authentication	Data Exchanged / Purpose
Sheet Sync	Zoho Sheet – ReadTrack Book Records, Sheet1	Connection + zoho.sheet.createRecords	OAuth	Book Title, Author, Genre and Status of each new book
Pivot Table	Zoho Analytics	Built-in integration	Platform account	Pages Read by Book Title
Email	Email (sendmail)	Email workflows	Platform account	Notification when a record is added

Code 5.5: Zoho Sheet synchronisation (DS-05)

```
sheetRecord = Map();
sheetRecord.put("Book Title", input.Book_Title);
sheetRecord.put("Author", input.Author);
sheetRecord.put("Genre", input.Genre);
sheetRecord.put("Status", input.Status);

queryData = Map();

response = zoho.sheet.createRecords(
    "zgq8i320bbc78e41b41fea63d7decb410dd95",
```

```
    "Sheet1",
    sheetRecord,
    queryData,
    "zoho_sheet_connection"
);

info response;
```

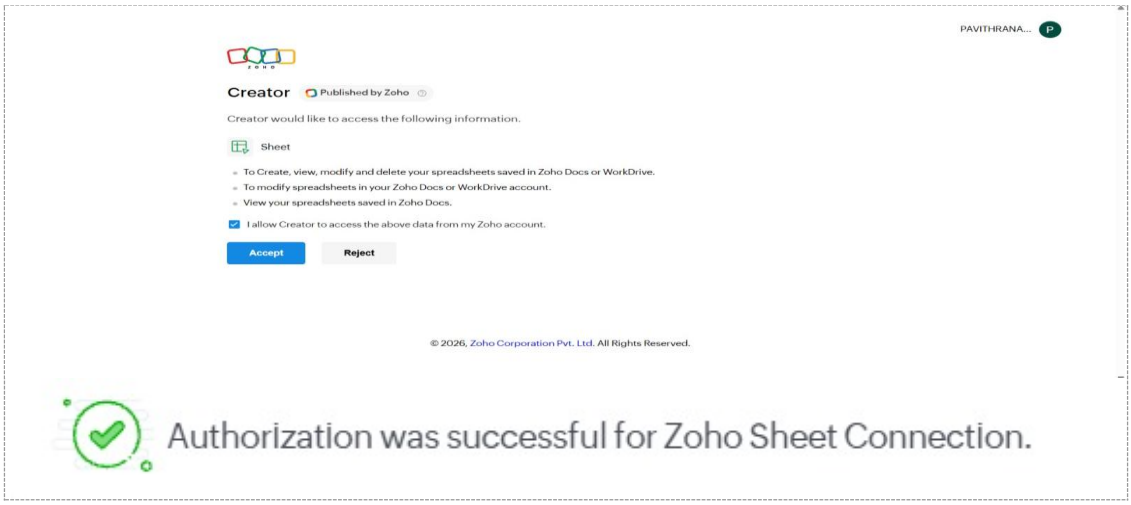


Figure 5.15: Zoho Sheet Connection Authorised in Zoho Creator

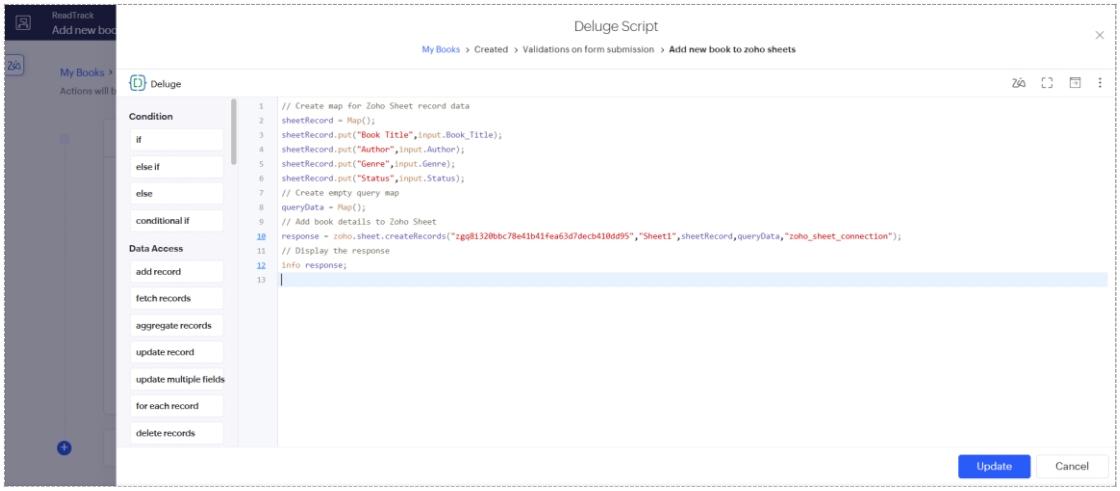


Figure 5.16: Zoho Sheet Synchronisation Script

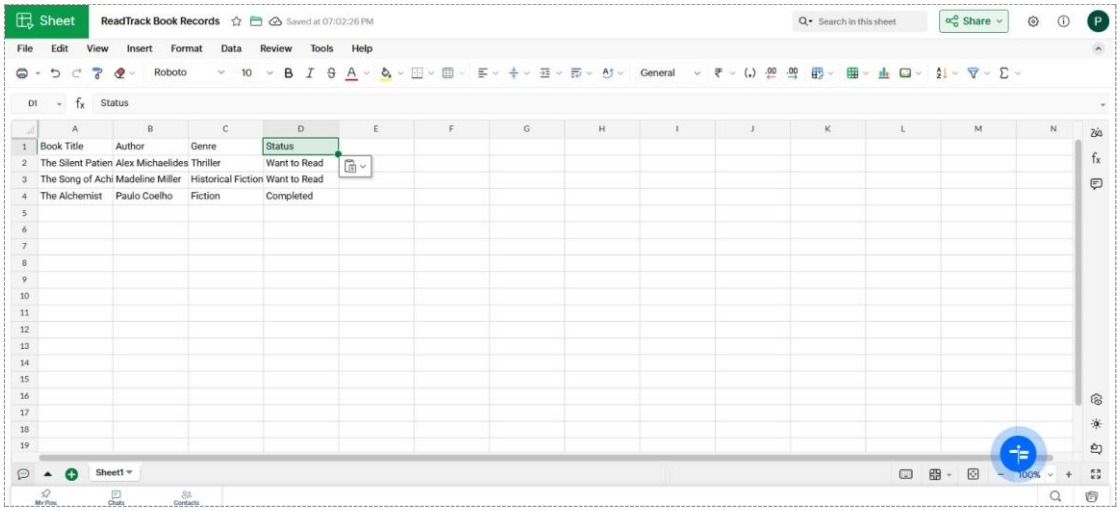


Figure 5.17: Synced Book Records in the Zoho Sheet

5.8 User Management and Role-Based Access

Two roles, **Admin** and **Reader**, and two permission sets, **Admin Access** and **Reader Access**, were created. Reader Access is configured for My Books, Reading Progress and Book Reviews with Access, View and Add permissions, without View All or Modify All. Three sharing rules (Table 5.6) share the records that Readers add with the Admin as Read Only. The dashboard and home page were hidden from the Reader view during permission configuration because the shared statistics were not suitable for readers.

Table 5.6: Sharing Rules

Rule Name	Form	Shared With	Access Level
Admin My Books Access	My Books	Admin (Reader records)	Read Only
Admin Reading Progress Access	Reading Progress	Admin (Reader records)	Read Only
Admin Book Review Access	Book Reviews	Admin (Reader records)	Read Only

Testing confirmed that the Admin can view a Reader's newly added book, that the Reader can add records, and that the Reader cannot view the Admin's original books under the configured access arrangement (Section 6.2).

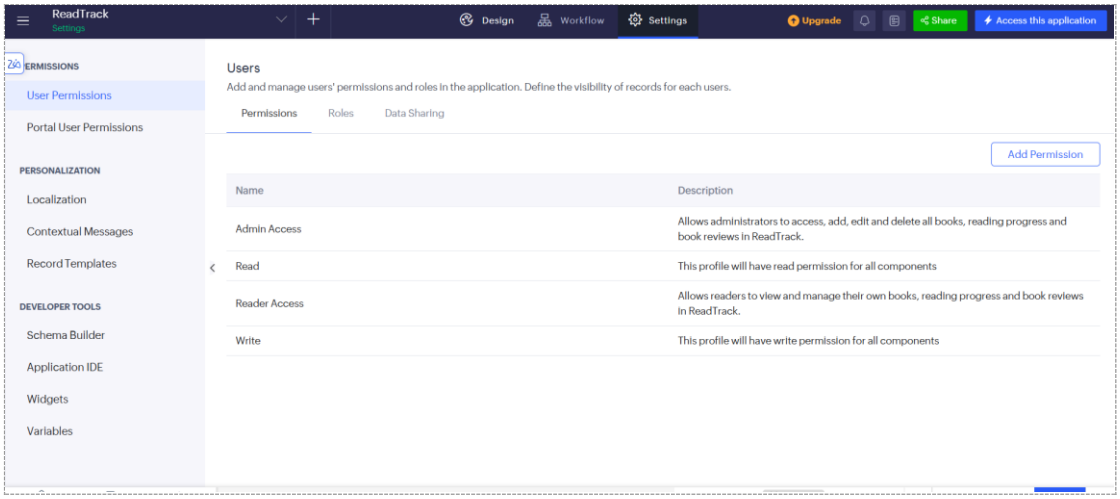


Figure 5.18: Roles and Permission Sets for Admin and Reader

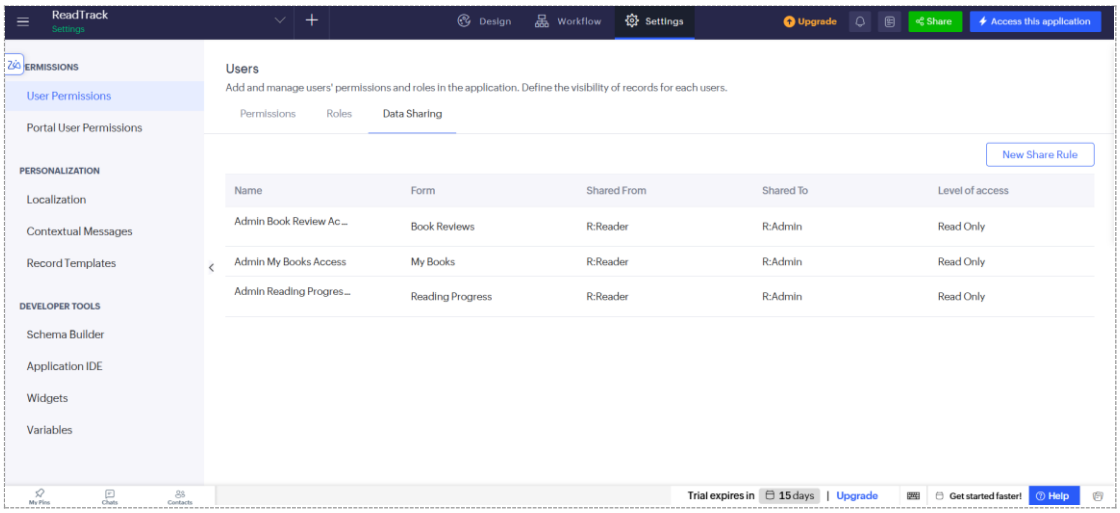


Figure 5.19: Sharing Rules

5.9 AI and Analytics Features

ReadTrack uses the analytics features built into Zoho Creator. The ReadTrack Dashboard shows book counts together with the Books by Genre and Books by Status charts, which are built from the form data (Figure 5.8). A pivot table created with Zoho Analytics shows Pages Read by Book Title (Figure 5.7), and the calendar report shows reading updates by date (Figure 5.6). The data set is small, so AI-based insights were not needed.

5.10 Pro-Code Extension

A custom widget, **ReadTrack Goal Tracker**, was built with HTML, CSS and JavaScript, uploaded through Zoho Creator Widgets and added to the ReadTrack Home Page. It has a page goal input, a pages read input, a progress percentage calculation, a

visual progress bar, an Update Progress button and dynamic progress messages. The progress is calculated as:

$$\text{Progress} = (\text{Pages Read} \div \text{Page Goal}) \times 100$$

The percentage is capped at 100%. The widget currently works independently, using its own input values; it is not connected to the Reading Progress records in the database. Connecting it is listed as future work.

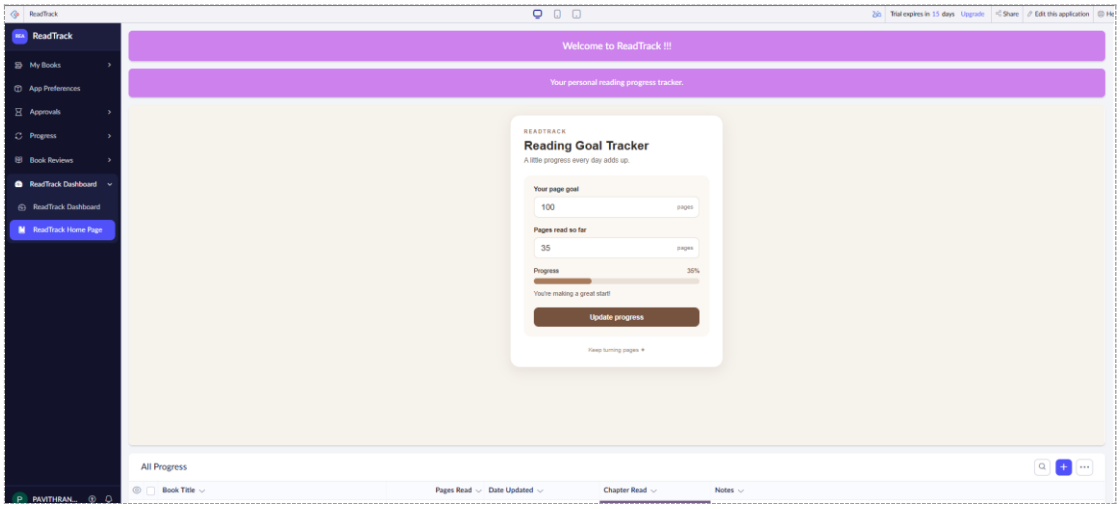


Figure 5.20: ReadTrack Goal Tracker Widget on the Home Page

5.11 Mobile and Portal Access

Portal access is available for ReadTrack, so users can sign in through a portal. The application was built and tested mainly in the web interface. Because Zoho Creator applications can be opened in the Zoho Creator mobile app, mobile use can be added later after testing; mobile-specific layouts were outside the scope of this project.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was done at four levels. **Unit testing** checked each form and each script with sample records. **Integration testing** checked that the workflows ran in the right order and that the Zoho Sheet connection created a record. **Role-based testing** checked what the Admin and Reader permission sets can see and do. **User acceptance testing** (Section 6.5) collects feedback from other users.

6.2 Test Cases

Table 6.1 gives 19 test cases that cover the functional requirements, including negative tests. All test cases passed.

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Add a book with all mandatory fields	Fill the five mandatory fields; submit	Record saved	Book saved with the five mandatory fields	Pass
TC-02	FR-01	Submit without Genre	Leave Genre empty	Mandatory-field error	Form not saved; mandatory-field error shown for Genre	Pass
TC-03	FR-04	Start date in the future	Enter tomorrow's date	Alert "Start Date cannot be in the future."; not saved	Alert "Start Date cannot be in the future." shown; record not saved	Pass
TC-04	FR-04	Start date today or earlier	Enter today's date	Record saved	Record saved	Pass
TC-05	FR-05	Pages Read is zero	Enter 0	Alert shown; submission cancelled	Alert shown; submission cancelled	Pass
TC-06	FR-05	Pages Read is negative	Enter -5	Alert shown; submission cancelled	Alert shown; submission cancelled	Pass
TC-07	FR-05	Date Updated in the future	Enter tomorrow's date	Alert shown; submission cancelled	Alert shown; submission cancelled	Pass
TC-08	FR-06	Date Updated left empty	Leave the date empty; submit	Date set to today	Date Updated filled with today's date	Pass
TC-09	FR-02, FR-07	Valid progress record	Select book, pages read 25	Record saved; progress email sent	Record saved; progress email received	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-10	FR-07	New book email	Add a new book	New Book Email received	New Book Email received (after the fix in BUG-01)	Pass
TC-11	FR-03, FR-08	Submit a review	Reader submits a review	Admin receives an approval notification	Admin received the approval notification	Pass
TC-12	FR-08	Reject a review	Admin rejects the review	Review is rejected	Review rejected by the Admin	Pass
TC-13	FR-09	Blueprint transition	Start Reading on a To Read book	Book moves to Currently Reading	Book moved to Currently Reading	Pass
TC-14	FR-10	Batch workflow	Run the batch workflow	Message "Reading Progress record processed successfully"	Message "Reading Progress record processed successfully" shown	Pass
TC-15	FR-11	Sync to Zoho Sheet	Add a new book	New row in ReadTrack Book Records (Sheet1)	New book record synchronised to the Zoho Sheet	Pass
TC-16	FR-13	Dashboard counts	Open dashboard after adding books	Counts match the records	Dashboard counts matched the records	Pass
TC-17	FR-14	Admin views a Reader's book	Reader adds a book; Admin opens My Books	Admin can view the new book (read-only)	Admin could view the Reader's newly added book	Pass
TC-18	FR-14	Reader adds records	Log in as Reader; add a record	Record saved	Reader could add records	Pass
TC-19	FR-14	Reader views Admin's books	Log in as Reader; open My Books	Admin's original books not visible	Reader could not view the Admin's original books	Pass

6.3 Debugging Deluge Scripts

Errors were found in four ways: (1) reading the error message shown on submission; (2) adding **info** statements to print variable values; (3) running the script from the Deluge editor; and (4) checking the workflow and connection settings. For example, when the confirmation email for a new book did not arrive, the workflows of each form were checked and the email script was found on the Book Reviews form instead of the My Books form; moving it to My Books fixed the problem (Table 6.2). For the Zoho Sheet script, the connection name, the sheet identifier, the worksheet name and the field names sent were checked, and the response was printed with an info statement.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	New-book confirmation email did not arrive when a book was added	The email script was attached to the Book Reviews form instead of the My Books form	Script moved to the My Books form and re-tested	Closed

6.5 User Acceptance Testing

The application was tried in both roles, Reader and Admin, using the demo data, and worked as expected.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Pavithra Narayanan (self-test)	Reader	Added books, progress entries and reviews; validations and email confirmations worked as expected	No change needed
Pavithra Narayanan (self-test)	Admin	Review approval, reports, dashboard and read-only view of reader records worked as expected	No change needed

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator supports separate Development, Stage and Production environments. ReadTrack was built and tested in the Development environment and then published, and it is live at the application URL given in the Project at a Glance table. If a problem is found after release, the fix is made and tested before the application is published again. The deployment of ReadTrack was completed.

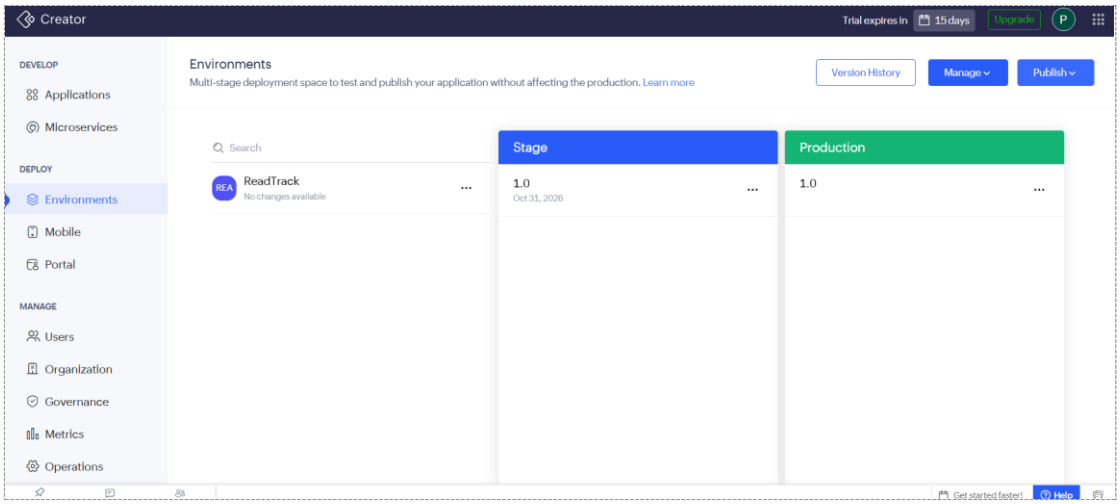


Figure 7.1: Environments / Publish Application

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development	✓
2	Sample/test data removed or isolated	✓
3	Roles, permissions and users configured in the live application	✓
4	Connections authorised in the live application	✓
5	Workflows enabled	✓
6	Application published and URL verified	✓
7	Version / release notes recorded	✓

7.2 Performance Optimisation

Performance was addressed in the design in several simple ways. The validation scripts are lightweight field checks that do not fetch other records. The batch workflow

processes Reading Progress records in batches of up to 1000. The All My Books report uses a filter and grouping so that a smaller set of records is shown. The Zoho Sheet synchronisation sends only four fields and runs once for each new book. The Goal Tracker widget does its calculation in the browser. Table 7.2 summarises these optimisations.

Table 7.2: Performance Optimisations Applied

Optimisation Applied	Component	Benefit
Lightweight validation scripts without record fetches	DS-01 to DS-04	Checks run instantly on the form
Batch processing (1000 records per batch)	Reading Progress batch workflow	Large sets are processed in manageable groups
Filter and grouping in the report	All My Books	Fewer records to load and read
Only four fields sent to the sheet	Zoho Sheet sync (DS-05)	Small, quick update for each new book

7.3 Security Features

Security is handled through permission sets, sharing rules, an OAuth connection, an approval step and input validation. Readers work only with their own records, and the dashboard and home page are hidden from them.

Table 7.3: Security Controls Implemented

Security Control	**Where Applied**	**Threat Mitigated**
Permission sets (Admin, Reader)	Settings → User Permissions	Unauthorised viewing, editing or deletion
Sharing rules (Read Only)	Settings → Data Sharing	Admin changing reader records by mistake
Dashboard and home page hidden	Reader view	Exposure of shared statistics
Review approval process	Book Reviews form	Unchecked reviews being accepted
OAuth connection	Zoho Sheet connection	Credential leakage
Validation scripts	My Books, Reading Progress	Invalid data entry

7.4 Governance and Administration

Good practice for this application includes: using clear form and field names; naming workflows after their purpose; recording version notes when publishing; reviewing user access regularly; and exporting data from reports or the Zoho Sheet as a backup. The

administrator is responsible for adding users, assigning permission sets, approving reviews and keeping the Zoho Sheet connection authorised. The Audit Trail was opened and the recorded changes were viewed (Figure 7.2).

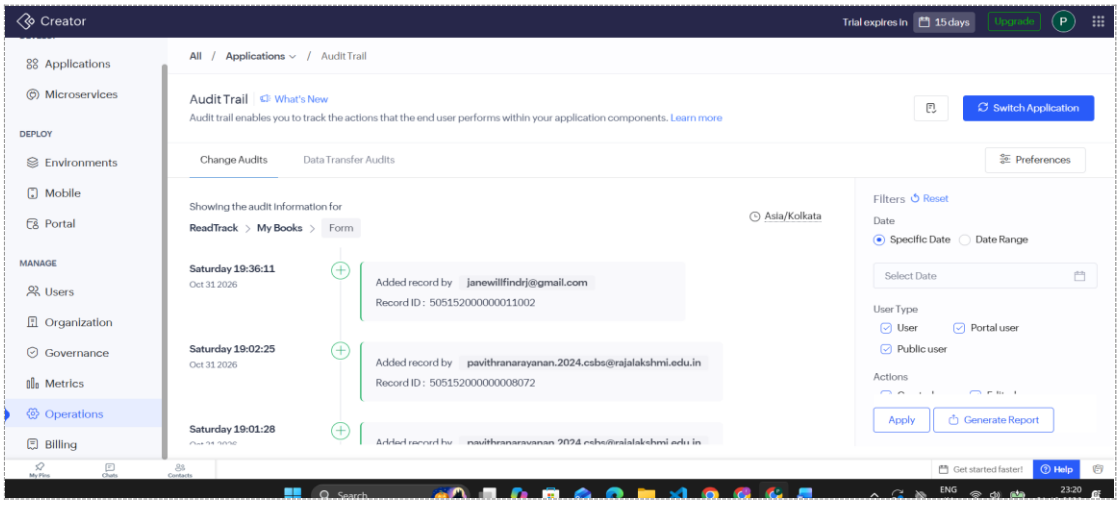


Figure 7.2: Audit Trail

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

One realistic scenario shows the application end to end with the demo data (about 20 books, 10 progress entries and 6 reviews). **Step 1:** a Reader adds a new book in My Books; the New Book Email is sent (Figure 8.1). **Step 2:** the book is copied to the Zoho Sheet ReadTrack Book Records (Figure 8.2). **Step 3:** the Reader submits a review, and the Admin receives the approval notification and approves or rejects it (Figure 8.3).

Book Title	**The Cruel Prince**
Author	Holly Black
Genre	Fantasy
Start Date	01-Nov-2026
Status	Want to Read
IBSN	9780316310277
Personal Notes	A fantasy story about a mortal girl navigating power, politics and betrayal in the world of faeries.

Figure 8.1: Scenario Step 1 – Reader Adds a Book

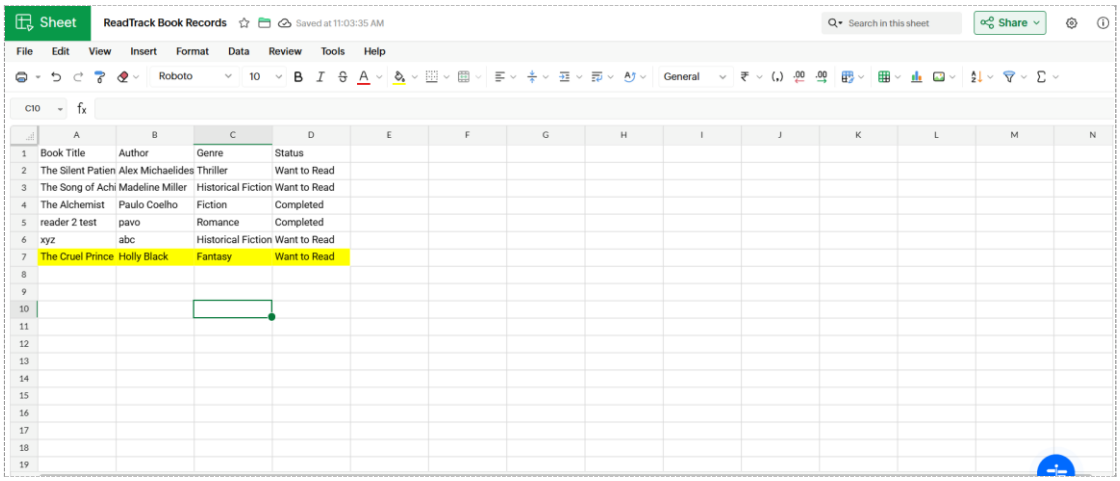


Figure 8.2: Scenario Step 2 – Book Record in the Zoho Sheet

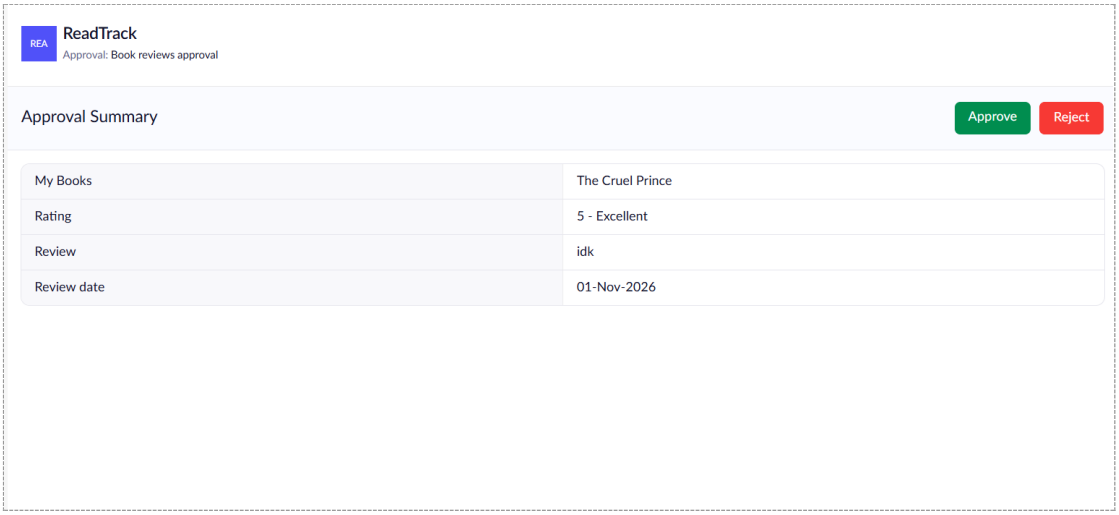


Figure 8.3: Scenario Step 3 – Admin Approval of a Review

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
1. Data model and forms	Sections 4.2, 4.3, 5.2	Three related forms and a subform (Figures 5.2–5.4)
2. Data checks, automatic entry, emails	Sections 5.5, 5.6	Validation scripts, automatic date entry and three email workflows (Figures 5.10, 5.11)
3. Reports, dashboard and home page	Sections 5.3, 5.4	Three list reports, calendar report, pivot table, dashboard and home page (Figures 5.5–5.9)
4. Blueprint and approval	Section 5.6	Book Reading Status Flow and review approval (Figures 5.12, 5.13)
5. Zoho Sheet integration	Section 5.7	New books synchronised to the sheet (Figures 5.15–5.17)
6. Access control and widget	Sections 5.8, 5.10	Permission sets, sharing rules and the Goal Tracker widget (Figures 5.18, 5.19, 5.20)
7. Testing, deployment and governance	Chapters 6 and 7	Test cases, security controls, deployment and audit trail

8.3 Limitations of the Low-Code Approach

ReadTrack meets its goals comfortably within what Zoho Creator offers. As with any low-code project, a few boundaries are worth noting, and each has a simple way forward. The interface follows the form builder, page builder and widget options, which was enough for this application. Deluge keeps the scripts short and readable. The Goal Tracker widget currently works on its own input values, and connecting it to the stored Reading Progress records is a natural next step. The application runs on the Zoho

platform, which also provides its hosting, security and integrations. These points agree with the general advice that low-code suitability should be checked for each project [4], and for ReadTrack the approach was a good fit.

8.4 Reflection on the No-Code, Low-Code, Pro-Code Spectrum

No-code: creating the forms, subform, lookups, reports, dashboard, approval process, blueprint, batch workflow, permission sets and sharing rules by configuration. **Low-code:** the Deluge scripts for validation, automatic date entry and the Zoho Sheet synchronisation. **Pro-code:** the Goal Tracker widget written in HTML, CSS and JavaScript. A larger move to pro-code would be justified if ReadTrack needed heavy analytics, a fully custom interface or large-scale integration with external book catalogues.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

ReadTrack was built on Zoho Creator to keep books, reading sessions, progress and reviews in related forms. Deluge scripts validate the start date, pages read and update date and fill the update date automatically; email workflows confirm new records; an approval process and a blueprint control reviews and book status; and a batch workflow processes progress records. Reports, a calendar report, a pivot table, a dashboard and a home page present the data, and a custom widget shows progress against a goal. New books are synchronised to a Zoho Sheet, and Admin and Reader permission sets with sharing rules control access.

The project shows that a small but complete business application, with automation, approval, integration and access control, can be built on a low-code platform.

9.2 Future Enhancements

- Connect the Goal Tracker widget to the Reading Progress records so that the percentage is calculated from stored data.
- Add a scheduled reminder for the next reading session and a calendar integration.
- Fetch book details such as title, author and cover from an external book API.
- Provide a mobile-friendly layout and test the Zoho Creator mobile app.

9.3 Learning Outcomes and Career Relevance

This project gave practice in data modelling, forms and subforms, reports, dashboards, Deluge scripting, workflows, approvals, blueprints, integration through connections, a custom widget, role-based access and testing. These skills relate to roles such as citizen developer, low-code developer, solutions consultant and automation engineer.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Y = implemented, P = partly implemented, N = not implemented.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	ReadTrack menu	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	Single Line, Multi Line, Dropdown, Date, Number, Checkbox, URL, Lookup	4.3	Fig. 5.2–5.4
II	Lookup field (relationship between forms)	Y	Book field in Reading Progress and Book Reviews	4.2	Fig. 5.3, 5.4
II	Subform (parent–child data)	Y	Reading Session in My Books	4.2, 5.2	Fig. 5.2
II	Field validations / layout rules	Y	Mandatory fields; start date, pages read and date updated checks	5.5	Code 5.1–5.3
II	List report with filter, grouping, custom action	Y	All My Books (Genre = Fantasy, grouped by Author, custom action)	5.3	Fig. 5.5
II	Kanban / Calendar / Timeline / Map report	Y	Calendar report from Reading Progress	5.3	Fig. 5.6
II	Pivot table / chart report	Y	Pivot table (Zoho Analytics); dashboard charts	5.3, 5.4	Fig. 5.7, 5.8
II	Page / dashboard built with UI Builder	Y	ReadTrack Dashboard and Home Page	5.4	Fig. 5.8, 5.9
II	Deluge – form events (validate / success)	Y	DS-01 to DS-05	5.5	Code 5.1–5.5
II	Deluge – functions, fetch, loops, collections	P	Map and conditions used in the scripts; no standalone function or fetch	5.7	Code 5.5
II	Email / in-app notifications	Y	New Book, Reading Progress and Book Review emails	5.6	Fig. 5.11
II	Form workflow	Y	Validation, automatic entry and Zoho Sheet sync	5.6	Fig. 5.10
II	Scheduled workflow	N	—	5.6	—
II	Approval process	Y	Book Reviews	5.6	Fig. 5.12
II	Blueprint	Y	Book Reading Status Flow	5.6	Fig. 5.13
II	Batch / report workflow	N	—	5.6	—

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Testing & debugging evidence	Y	Test cases and debug screenshots	6	Table 6.1, 6.2
III	Integration with a Zoho service	Y	Zoho Sheet (working); Zoho Analytics	5.7	Fig. 5.15–5.17
III	Connection / REST API call	Y	zoho_sheet_connection with zoho.sheet.createRecords	5.7	Code 5.5
III	Webhook or Creator REST API	N	—	5.7	—
III	Users, roles and permission profiles	Y	Admin and Reader roles; Admin Access and Reader Access	5.8	Fig. 5.18
III	Record-level access criteria / customer portal	Y	Sharing rules; portal access available	5.8, 5.11	Fig. 5.19
III	Zia / AI feature	Y	Zia used for suggestions and help with some Deluge scripts	5.5	—
III	Analytics (charts / Zoho Analytics)	Y	Dashboard charts; Zoho Analytics pivot	5.9	Fig. 5.7, 5.8
IV	Performance optimisation applied	P	Applied in design (Table 7.2)	7.2	Table 7.2
IV	Deployment across environments	Y	Built in Development and published as a live application	7.1	Fig. 7.1
IV	Security controls (OAuth, permissions)	Y	Permission sets, sharing rules, OAuth connection, approval	7.3	Table 7.3
IV	Governance (audit trail, administration)	Y	Audit Trail viewed	7.4	Fig. 7.2
V	Pro-code extension (widgets / Catalyst / functions)	P	ReadTrack Goal Tracker widget (HTML, CSS, JavaScript); not connected to the database	5.10	Fig. 5.20
V	Low-code limitations & spectrum analysis	Y	Discussion	8.3, 8.4	—
—	Mobile app / portal access	Y	Portal access available;	5.11	—

Total features implemented: 26 / 32 (partly implemented: 3)

APPENDIX B – COURSE OUTCOME MAPPING

Self-assess the attainment level (1 – Low, 2 – Medium, 3 – High); the examiner may revise it.

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7, 5.8	3	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	2	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.3, 8.4, 9.3	2	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 My Books – Start Date Validation (DS-01)

```
if(input.Start_Date > zoho.currentdate)
{
    alert "Start Date cannot be in the future.";
    cancel submit;
}
```

C.2 Reading Progress – Pages Read Validation (DS-02)

```
if(input.Pages_Read <= 0)
{
    alert "Pages Read must be greater than zero. Please enter a
valid positive number.";
    cancel submit;
}
```

C.3 Reading Progress – Date Updated Validation (DS-03)

```
if(input.Date_Updated > zoho.currentdate)
{
    alert "Date Updated cannot be in the future. Please select
today or an earlier date.";
    cancel submit;
}
```

C.4 Reading Progress – Automatic Date Entry (DS-04)

```
if(input.Date_Updated.isNull())
{
    input.Date_Updated = zoho.currentdate;
}
```

C.5 My Books – Zoho Sheet Synchronisation (DS-05)

```
sheetRecord = Map();
sheetRecord.put("Book Title", input.Book_Title);
sheetRecord.put("Author", input.Author);
sheetRecord.put("Genre", input.Genre);
sheetRecord.put("Status", input.Status);

queryData = Map();

response = zoho.sheet.createRecords(
    "zgq8i320bbc78e41b41fea63d7decb410dd95",
    "Sheet1",
    sheetRecord,
    queryData,
    "zoho_sheet_connection"
);

info response;
```

C.6 Reading Progress – Batch Workflow Script

```
info "Reading Progress record processed successfully";
```

C.7 Email Workflows (New Book)

```
userEmail = zoho.loginuserid;
sendmail
[
      from :zoho.adminuserid
      to :userEmail
      subject :"New Book Review Submitted"
      message :"A new book review has been submitted successfully
to ReadTrack.Thank you for sharing your reading experience!"
]
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the ReadTrack application link https://readtrack.zohocreatorportal.in in a web browser, or sign in through the portal, using your Zoho account. Admin users see the home page and dashboard. Reader users see the forms and reports allowed for the Reader role.

D.2 Instructions for the Administrator

1. Open Settings → User Permissions to check the Admin Access and Reader Access permission sets.
2. Open Data Sharing to check the three sharing rules that give you read-only access to reader records.
3. Open the approval notification for a submitted review and choose Approve or Reject.
4. Open the dashboard and the home page to see book counts, charts and the Goal Tracker widget.
5. Check the Zoho Sheet connection and re-authorise it if needed; open the Audit Trail to review changes.

D.3 Instructions for the Reader

1. Open My Books and click Add to enter a new book. Book Title, Author, Genre, Start Date and Status are required, and the start date cannot be in the future.
2. Add reading sessions in the Reading Session subform if needed.
3. Open Reading Progress, choose the book and enter the pages read (greater than zero), the date and notes. If you leave the date empty, today's date is used.
4. Open Book Reviews to add a rating and a review; the Admin approves or rejects it.
5. Use the All My Books, All Progress and All Book Reviews reports to see your own records.